

2.Requirement Analysis Phase

Project Title

Script-Controlled ACL – Restrict Record Access Based on Field Value

Functional Requirements

The system should control access to records based on a specific field value. The ACL script should check the required field before allowing the user to access the record.

The main functional requirements are:

- Identify the user requesting the record.
- Read the selected field value from the record.
- Compare the field value with the defined condition.
- Allow access when the condition is satisfied.
- Restrict access when the condition is not satisfied.
- Apply the access rule automatically whenever the record is requested.

Non-Functional Requirements

Security:

The system should prevent unauthorized users from accessing restricted records.

Reliability:

The ACL should provide a consistent access decision for different users and records.

Performance:

The script should execute efficiently without causing unnecessary delay.

Maintainability:

The ACL script should be simple to understand and easy to modify when requirements change.

Software Requirements

The following software is required:

- ServiceNow: Used to create and configure the ACL.
- JavaScript: Used to write the access-control script.
- Web Browser: Used to access and test the ServiceNow instance.
- ServiceNow Account: Required for configuring and testing the project.

System Requirements

The project requires a basic computer or laptop with:

- Stable internet connection
- Modern web browser
- ServiceNow instance
- Valid ServiceNow account
- Required permissions to configure ACLs

Since the project is implemented using the ServiceNow platform, no special hardware is required.

User Requirements

Administrator:

The administrator creates and configures the ACL and defines the field-based access condition.

Authorized User:

The authorized user can access the record when the required field condition is satisfied.

Unauthorized User:

The unauthorized user should not be able to access a record when the required condition is not satisfied.

ACL Requirements

The ACL is the main security component of the project. It should be configured for the required table and operation.

The script checks the selected field value and returns an access decision.

Example:

Matching Field Value → Access Allowed

Non-Matching Field Value → Access Restricted

For example, if the Department field is used, the system can compare the department of the record with the department associated with the user.

Expected Result

The expected result of the requirement analysis is to clearly identify all requirements needed to implement the Script-Controlled ACL.

The final access flow should be:

User Request → ACL Check → Field Value Verification → Access Allowed / Restricted

This ensures that records are accessed only when the defined condition is satisfied.

Conclusion

The Requirement Analysis Phase defines the functional, technical, user, and security requirements of the project. It establishes how the ACL will check field values and control record access.

These requirements provide the foundation for designing and implementing the Script-Controlled ACL in the following project phases.